



The weighted Ehrhart theory of graphical zonotopes

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1. (Weighted) Ehrhart Theory

2. Graphical Zonotopes

3. Results

 Computations

 Rank Conjecture

 Future Questions



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An introduction to polytopes



A (integral) polytope $P \subseteq \mathbb{R}^d$ is defined as the convex hull of a finite set of points, equivalently, the closed bounded intersection of finitely many half-spaces. Equivalently, P is the solution set to $Ax \leq \mathbf{b}$ for some (integral) matrix $A \in \mathbb{Z}^{n \times d}$ and (integral) vector $\mathbf{b} \in \mathbb{Z}^n$.

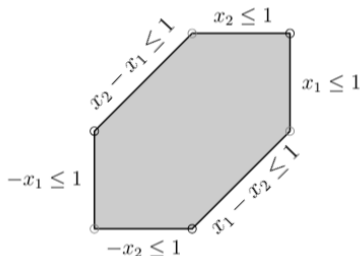


Figure: A example of an polytope. Modified from Joswig and Kulas [2010].

Ehrhart Theory (1)



"The central problem of Ehrhart Theory is counting the number of **integer points** in **integer dilates** of **integral polytopes**."

Integral polytopes are polytopes with all integral vertices.

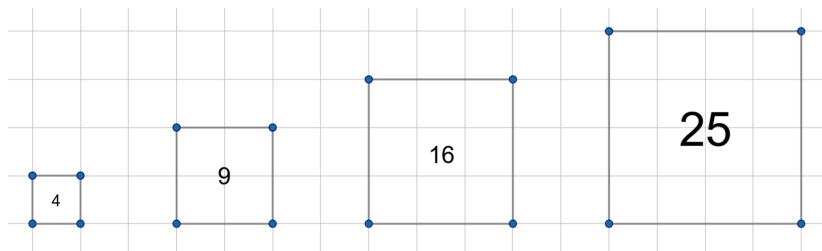


Figure: The first four integer dilates of the square P with vertices $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$ and their corresponding integer point counts.

Question for the audience: In this case, what is $\#(tP \cap \mathbb{Z}^d)$?

Theorem (Ehrhart Polynomial, Ehrhart [1967])

For any integral d -dimensional polytope P , $\text{ehr}_P(t) = \#(tP \cap \mathbb{Z}^d)$ is a degree d polynomial, which we refer to as the **Ehrhart polynomial**.

Definition (Ehrhart Series)

Moreover, if we encode the numbers $\text{ehr}_P(0), \text{ehr}_P(1), \text{ehr}_P(2), \dots$ as the coefficients of a power series, we get the **Ehrhart series**

$$\text{Ehr}_P(z) = \sum_{t \geq 0} \text{ehr}_P(t) z^t = \frac{h_d^* z^d + \dots + h_0^*}{(1 - z)^{d+1}}$$

We refer to the numerator $h_P^*(z) = h_d^* z^d + \dots + h_0^*$ as the **h^* -polynomial**.

Ehrhart Theory (2.5)

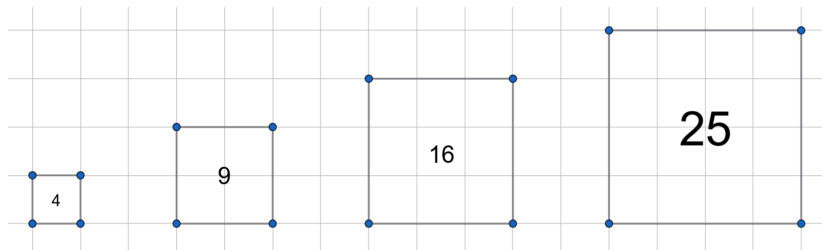


Figure: The Ehrhart polynomial of P in this case is $\text{ehr}_P(t) = (t + 1)^2$ and the corresponding Ehrhart series is $\text{Ehr}_P(z) = 1 + 4z + 9z^2 + 16z^3 + 25z^4 + \dots = (z + 1)/(1 - z)^3$.

Weighted Ehrhart Theory (1)



Now what happens if we “weigh” each of our points by a polynomial function?

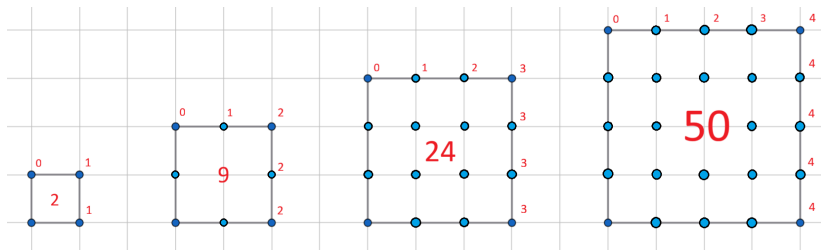


Figure: The first four integer dilates of the square P with vertices $(0,0)$, $(0,1)$, $(1,0)$, $(1,1)$ and their corresponding **weighted** integer point counts according to the function $w = x_1$.

Weighted Ehrhart Theory (2)



Definition (Weighted Ehrhart Polynomial)

For any integral d -dimensional polytope P and homogeneous polynomial w of degree m , $\text{ehr}_{P,w}(t) = \sum_{p \in tP \cap \mathbb{Z}^d} w(p)$ is a degree $d + m$ polynomial, called the **weighted Ehrhart polynomial**.

Definition (Weighted Ehrhart Series)

Moreover, if we encode $\text{ehr}_{P,w}(0), \text{ehr}_{P,w}(1), \text{ehr}_{P,w}(2), \dots$ as the coefficients of a power series, we get the **weighted Ehrhart series**

$$\text{Ehr}_{P,w}(z) = \sum_{t \geq 0} \text{ehr}_{P,w}(t) z^t = \frac{h_{d+m}^* z^{d+m} + \dots + h_0^*}{(1-z)^{d+m+1}}$$

We refer to the numerator $h_{P,w}^*(z) = h_{d+m}^* z^{d+m} + \dots + h_0^*$ as the **weighted h^* -polynomial**.

Weighted Ehrhart Theory (2.5)



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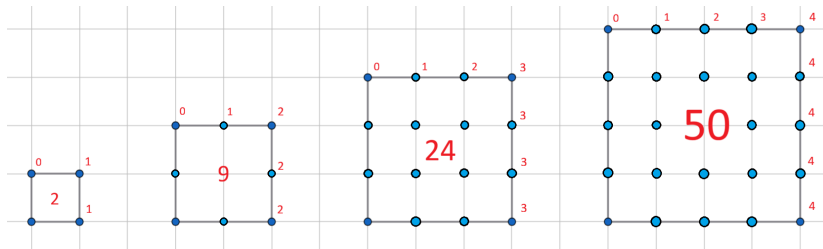


Figure: The weighted Ehrhart polynomial of P and w in this case is $\text{ehr}_{P,w}(t) = (t+1) \sum_{x_1=0}^t x_1 = \frac{1}{2}t(t+1)^2$

and the corresponding weighted Ehrhart series is

$$\text{Ehr}_{P,w}(z) = 2z + 9z^2 + 24z^3 + 50z^4 + \dots = (z^2 + 2z)/(1-z)^4.$$

Theorem (Stanley [1980])

For any d -dimensional integral polytope P , $h_P^(z)$ has nonnegative integer coefficients.*

However, this is not true in the weighted case! Not even for specialized cases of polytopes and weights!

Original Motivating Question

Which combinations of integral polytopes P and homogeneous weight functions w have weighted h^* -polynomials $h_{P,w}^*(z)$ with nonnegative coefficients?

Today's question: What happens when P is a graphical zonotope?



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Definition (Graphical zonotopes)

Let G be a finite simple connected graph with $d + 1$ vertices. The **graphical zonotope** Z_G is defined as the **Minkowski sum** of the vectors $e_j - e_i$ for each edge $(i, j) \in E(G)$ (with $i < j$).

In mathematical notation,

$$Z_G = \{c_{ij}(e_j - e_i) \mid 0 \leq c_{ij} \leq 1\} \subseteq \mathbb{R}^{d+1}$$

Example

Let $G = P_3$, the path graph on 3 vertices. Then

$$Z_G = [0, e_2 - e_1] + [0, e_3 - e_2].$$

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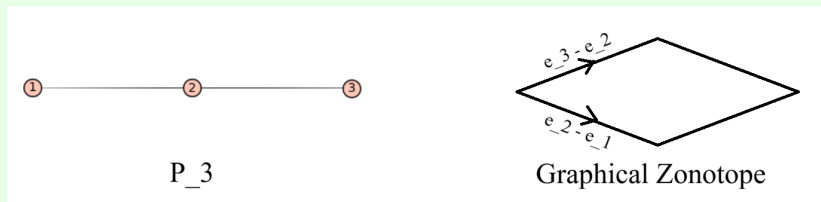


Figure: The path graph P_3 and its corresponding graphical zonotope Z_{P_3} .

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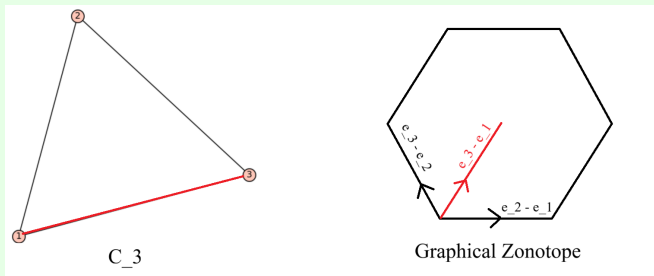


Figure: The cycle graph C_3 and its corresponding graphical zonotope Z_{C_3} .

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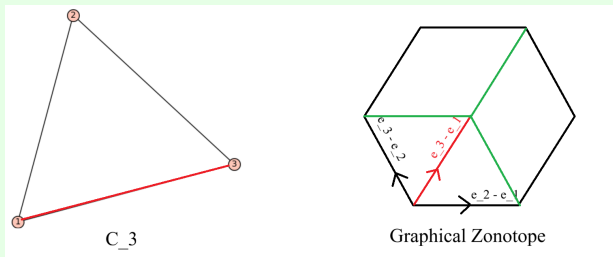


Figure: The cycle graph C_3 and its corresponding graphical zonotope Z_{C_3} ... it can be tiled into parallelepipeds (separated by green lines).

Theorem (Shephard [1974], Beck, Jochemko, and McCullough [2019])

Every zonotope has a decomposition into (half-open) parallelepipeds.

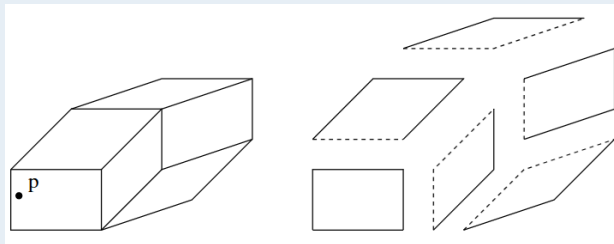


Figure: A decomposition of a zonotope into half-open parallelepipeds. Credits to Beck, Jochemko, and McCullough [2019].

We introduce a generalization of the Eulerian polynomial:

Definition $((A, j)$ -Eulerian polynomial)

Let $j \in \{1, \dots, d\}$.

We define the (A, j) -**Eulerian number** as

$$a_j(d, k) = \#\{\sigma \in S_d \mid \sigma_d = d + 1 - j \text{ and } \text{des}(\sigma) = k\}$$

and the (A, j) -**Eulerian polynomial** as

$$A_j(d, z) = \sum_{k=0}^{d-1} a_j(d, k) z^k.$$

Definition ((A,j)-Eulerian polynomial)

The (A,j) -**Eulerian polynomial** is defined as

$$A_j(d, z) = \sum_{k=0}^{d-1} a_j(d, k) z^k.$$

Example

$$A_1(3, z) = 1 + z$$

123, 213

$$A_2(3, z) = 2z$$

132, 312

$$A_3(3, z) = z^2 + z$$

321, 231

Theorem (Beck, Jochemko, and McCullough [2019])

Let C_j^d be the cube $[0, 1]^d$ missing j facets $x_d = 1, \dots, x_{d+1-j} = 1$.
Then

$$\text{Ehr}_{C_j^d}(z) = \frac{A_{j+1}(d+1, z)}{(1-z)^{d+1}}$$

Example

$$\text{Ehr}_{C_1^2}(z) = \frac{A_2(3, z)}{(1-z)^3} = \frac{2z}{(1-z)^3}$$

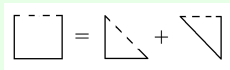


Figure: Decomposition of C_1^2 into half-open simplices.

We can generalize this to half-open parallelepipeds.

Theorem (Beck, Jochemko, and McCullough [2019])

The h^ -polynomial of a zonotope Z can be written as*

$$h_Z^*(z) = \sum_{j=0}^d c_{j+1} A_{j+1}(d+1, z)$$

with $c_1, \dots, c_{d+1} \geq 0$. Moreover, it is real-rooted and thus its coefficients are unimodal.

Specializing the previous result to graphical zonotopes, we get

Corollary (Mitukiewicz [2024])

The h^ -polynomial of a graphical zonotope Z_G can be written as*

$$h_{Z_G}^*(z) = \sum_{j=0}^d c_{j+1} A_{j+1}(d+1, z)$$

where c_{j+1} is the number of spanning forests of G with j internally passive edges.

Moreover, we have an explicit zonotopal subdivision for a graphical zonotope.

Theorem (Backman, Baker, and Yuen [2019])

Let A be a totally unimodular matrix $\mathbb{R}^{d \times n}$ and Z_A be the zonotope defined by the Minkowski sum of the columns of A . Then there exists an explicit zonotopal subdivision of Z_A into

$$Z(B) = \sum_{e \in B} [0, A_e] + \sum_{e \notin B} v_e$$

where B iterates over all bases of A .



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The results in Beck, Jochemko, and McCullough [2019] assume **translation-invariance**. However, **weighted Ehrhart polynomials are NOT translation-invariant**.

This makes weighted Ehrhart polynomials difficult to compute, even for simple graph cases.

In this presentation, we discuss the path graph and the cycle graph.

Lemma (H. (2026+))

For any zonotope Z and integral unimodular transformation $A \in \mathbb{Z}^{(d+1) \times (d+1)}$, we have

$$h_{Z,w(Ax)}^* = h_{A \cdot Z, w(x)}^*$$

by change of coordinates.

Proof.

Zonotopes are IDP and thus it suffices to show that A induces a bijection between the generators of Z and $A \cdot Z$. This is true since A is integral unimodular. $\square$

Corollary (H. (2026+))

Let P_{d+1} be the path graph with $d + 1$ vertices. It follows that

$$h_{C_d, w(Ax)}^*(z) = h_{Z_{P_{d+1}}, w}^*(z)$$

by the matrix

$$A = \begin{pmatrix} 1 & & & & & \\ -1 & \ddots & & & & \\ & \ddots & \ddots & & & \\ & & \ddots & \ddots & & \\ & & & \ddots & \ddots & \\ & & & & -1 & 1 \end{pmatrix}$$

where C_d is the d -dimensional cube.

Note A sends $e_i \rightarrow e_i - e_{i+1}$ for $1 \leq i \leq d$.

Reframing Path Graph (2.5)



Lemma (H. (2026+))

Let C_d be the d -dimensional cube and let $w = x_1^{m_1} \dots x_d^{m_d}$. Then

$$\text{ehr}_{C_d, w}(t) = (1^{m_1} + \dots + t^{m_1}) \dots (1^{m_d} + \dots + t^{m_d})$$

Definition

Let $\mathcal{W} : \mathbb{R}[t] \rightarrow \mathbb{R}[z]$ be defined by the map

$$f(t) \rightarrow \frac{\mathcal{W}(f)(z)}{(1-z)^{\deg(f)}}$$

In other words, $\mathcal{W}(\text{ehr}_{P, w}(t)) = h_{P, w}^*(z)$.

Weighted Ehrhart of Cube (2)



Let's start off with a simple result...

Lemma (H. (2026+))

*For $p > 0$, we have $\mathcal{W}(1^p + \dots + t^p) = z * A_p(z)$.*

Proof.

We know that

$$\text{Ehr}_{C_p}(z) = \frac{A_p(z)}{(1-z)^{p+1}} = 1 + 2^p * z + 3^p * z^2 + \dots$$

Multiplying both sides by $\frac{z}{1-z}$ yields

$$\frac{z * A_p(z)}{(1-z)^{p+2}} = 1^p * z + (1^p + 2^p) * z^2 + \dots$$



Weighted Ehrhart of Cube (3)



Things get really complicated when we introduce **products** of sums of powers!

Lemma (H. (2026+))

For $p \geq 0$, we have

$$\mathcal{W}((1^p + \dots + t^p)(1^0 + \dots + t^0)) = z(A_{p+1}(z) + A_p(z)).$$

Lemma (H. (2026+))

For $p \geq 0$, we have $\mathcal{W}((1^p + \dots + t^p)(1^1 + \dots + t^1)) =$

$$\frac{1}{2}(zA_{p+2}(z) + (z^2 + z)A_{p+1}(z) + 2z^2A_p(z)).$$

And it's hard to compute things from here...

Big Question: Is there a general formula for the weighted h^* -polynomial of the cube?

Hadamard Product (1)



You may recognize this operation as the **Hadamard product** of two power series.

Definition (Hadamard Product)

Let $p(t)$ and $q(t)$ be two polynomials in t with power series $\sum_{t \geq 0} p(t)z^t = \frac{\mathcal{W}(p)}{(1-z)^{d_1+1}}$ and $\sum_{t \geq 0} q(t)z^t = \frac{\mathcal{W}(q)}{(1-z)^{d_2+1}}$ respectively. Then multiplying **entry-wise** the power series of $p(t)$ and $q(t)$ yields the **Hadamard product**.

$$\sum_{t \geq 0} p(t)q(t)z^t = \frac{\mathcal{W}(pq)}{(1-t)^{d_1+d_2+1}}$$

Note this is NOT the same thing as $\mathcal{W}(p) * \mathcal{W}(q)$!

Hadamard Product (2)



While the Hadamard product is hard to compute, it does preserve some nice properties.

Theorem (Wagner [1992])

If $\mathcal{W}(p)$ and $\mathcal{W}(q)$ are simple negative real rooted, so is $\mathcal{W}(pq)$.

Corollary (Davis, De Loera, Garber, Garzon Mora, Jochemko, Yu (2026+), H. (2026+))

Let C_d be the d -dimensional cube and let $w = x_1^{m_1} \dots x_d^{m_d}$ with all $m_i > 0$. Then $h_{C_d, w}^$ is simple negative real rooted.*

And it may tell us even more!

Theorem (Brändén, Ferroni, and Jochemko [2025])

If $\mathcal{W}(p)$ and $\mathcal{W}(q)$ are ultra log-concave, so is $\mathcal{W}(pq)$.

Finally, one more formula of interest:

Lemma (Folklore)

Faulhaber's formula states that

$$1^p + \dots + t^p = \frac{1}{p+1} \sum_{r=0}^p \binom{p+1}{r} B_r^+ t^{p+1-r}$$

where B_r^+ are the second Bernoulli numbers.

So for now, we can at least compute weighted Ehrhart polynomials of the cube by multiplying Faulhaber polynomials together.

Cycle Graph Case (1)



We now switch to the cycle graph C_{d+1} with $d + 1$ vertices. Recognize it is the same as the path graph with additional edge $e_{d+1} - e_1 = (e_2 - e_1) + \dots + (e_{d+1} - e_d)$.

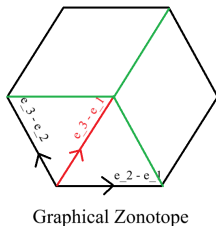
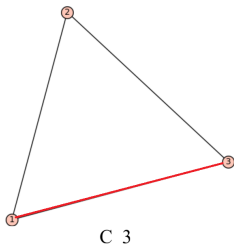


Figure: The cycle graph C_3 and its corresponding graphical zonotope Z_{C_3} .

So by a change of coordinates, we can investigate the weighted h^* -polynomial of $Z(e_1, \dots, e_d, e_1 + \dots + e_d)$ instead of $Z_{C_{d+1}}$.

Cycle Graph Case (2)



We can use the zonotopal subdivision described by Backman, Baker, and Yuen [2019], compute the weighted Ehrhart (and thus weighted h^* -) polynomial of each shifted half-open parallelepiped, and then add them together (in progress).

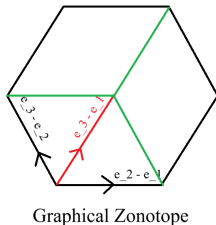
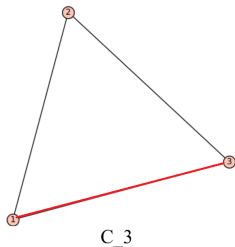


Figure: The cycle graph C_3 and its corresponding graphical zonotope Z_{C_3} .

Armed with our computational formulas, we compute the space of weighted h^* -polynomials for our graphical zonotopes Z_G .

Definition ($L_{Z_G, m}$)

Let G be a finite simple connected graph. Then define $w \rightarrow h_{Z_G, w}^*$ to be the map from all homogeneous weights of degree m to their corresponding weighted h^* -polynomials. Define the range of this map to be $L_{Z_G, m}$.

Motivating Question

What is the rank of $L_{Z_G, m}$?

Let $G = P_{d+1}$ be the path graph with $d + 1$ vertices.

	A	B	C	D	E	F	G	H	I
1	d down, m right	1	2	3	4	5	6	7	8
2	1	1	1	1	1	1	1	1	1
3	2	1	2	2	3	3	4	4	5
4	3	1	2	2	4	4	6	6	8
5	4	1	2	2	4	4	6	6	8
6	5	1	2	2	4	4	6	6	8
7	6	1	2	2	4	4	6	6	8
8	7	1	2	2	4	4	6	6	8
9	8	1	2	2	4	4	6	6	8
10	9	1	2	2	4	4	6	6	8

Figure: rank $L_{Z_{P_{d+1}}, m}$ for different d and m . Notice that the rank stabilizes as d increases.

Cycle Graph Experiments



Let $G = C_{d+1}$ be the cycle graph with $d + 1$ vertices.

	A	B	C	D	E	F	G	H	I
1	d down, m right	1	2	3	4	5	6	7	8
2	1	1	1	1	1	1	1	1	1
3	2	1	2	2	3	3	4	4	5
4	3	1	2	2	4	4	7	7	10
5	4	1	2	2	4	4	7	7	10
6	5	1	2	2	4	4	7	7	10
7	6	1	2	2	4	4	7	7	10
8	7	1	2	2	4	4	7	7	10
9	8	1	2	2	4	4	7	7	10
10	9	1	2	2	4	4	7	7	10
11	10	1	2	2	4	4	7	7	10
12	11	1	2	2	4	4	7	7	10

Figure: rank $L_{Z_{C_{d+1}}, m}$ for different d and m . Notice that the rank stabilizes as d increases.

Based on these observations, we make the following conjecture.

Conjecture (H. 2026+)

For sufficiently large d ,

$$\text{rank } L_{Z_{P_{d+1}},m} = \begin{cases} 2\lfloor \frac{m}{2} \rfloor & \text{if } m \geq 2 \\ 1 & \text{if } m = 1 \end{cases}$$

and

$$\text{rank } L_{Z_{C_{d+1}},m} = \begin{cases} 3\lfloor \frac{m}{2} \rfloor - 2 & \text{if } m \geq 2 \\ 2 & \text{if } m = 2, 3 \\ 1 & \text{if } m = 1 \end{cases}$$

We note that the right hand side does not depend on d .

What's next?



- Determining a formula for the weighted h^* -polynomials of shifted half-open cubes and parallelepipeds.
- Implementing an efficient algorithm to compute the weighted h^* -polynomial of a graphical zonotope.
- Resolving the rank conjecture.
- Proving rank $L_{P,2k} = \text{rank } L_{P,2k+1}$.
- Expanding our investigation to other families of graphs such as fan graphs and friendship graphs.

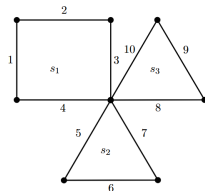
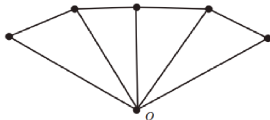


Figure: Examples of fan graphs and friendship graphs. Credits to Mitukiewicz [2024].

- We wish to compute weighted h^* -polynomials of graphical zonotopes Z_G .
- We have a nice interpretation for unweighted h^* -polynomials... but it does not (immediately) generalize to weighted h^* -polynomials.
- If G is the path graph, then Z_G is equivalent to the cube. Computing weighted h^* -polynomials on the cube is difficult.
- If G contains more edges than the path graph, we can subdivide Z_G into (half-open) parallelepipeds and compute each of their weighted h^* -polynomials.
- We wish to investigate the rank of the space of weighted h^* -polynomials of graphical zonotopes and investigate as $d \rightarrow \infty$ or $m \rightarrow \infty$.

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Thank you!



I am on the job market!
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